Suggested problems: 1, 3, 6, 8

Problem 1: Consider a sample space $S = \{a, b, c, d\}$ with probability measure P given by

s	$P(\{s\})$
a	0.1
b	0.2
c	0.3
d	0.4

Suppose we have three random variables X_1, X_2, and X_3 with values

s	$X_1(s)$	$X_2(s)$	$X_3(s)$
a	2	3	1
b	5	4	3
c	1	0	1
d	2	3	1

Describe the probability distribution of the 3-dimensional random vector $\mathbf{X} = (X_1, X_2, X_3)$.

Problem 2: Suppose that a machine has three parts, and part i will fail at time X_i, for $i = 1, 2, 3$. Suppose that the c.d.f. of $\mathbf{X} = (X_1, X_2, X_3)$ is

$$F(x_1, x_2, x_3) = \begin{cases} (1 - e^{-x_1})(1 - e^{-2x_2})(1 - e^{-3x_3}) & : x_1, x_2, x_3 \geq 0, \\ 0 & : \text{else.} \end{cases}$$

Compute the p.d.f. of $\mathbf{X}$.

Problem 3: Suppose that a point (X_1, X_2, X_3) is chosen uniformly at random from the set

$$S = \{(x_1, x_2, x_3) : 0 \leq x_i \leq 1,\ i = 1, 2, 3\}.$$

Compute the following:

(a) $P\left(\left(X_1 - \frac{1}{2}\right)^2 + \left(X_2 - \frac{1}{2}\right)^2 + \left(X_3 - \frac{1}{2}\right)^2 \leq \frac{1}{4}\right)$

(b) $P\left(X_1^2 + X_2^2 + X_3^2 \leq 1\right)$